Design and Evaluation of a Parameterised SIMD Vector Processing Unit on FPGA

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Repository: https://github.com/mahir-m01/SIMD-Vector-Processing-Unit

***Abstract*—Many workloads in signal processing, graphics and machine learning apply the same arithmetic to long arrays of numbers. Single-instruction, multiple-data (SIMD) hardware exploits this regularity by letting one instruction drive several arithmetic lanes at once. This report presents the Phase I design of a small, fully parameterised SIMD vector processing unit written in Verilog. It provides eight 128-bit vector registers split into eight 16-bit lanes, a 14-operation lane ALU replicated per lane, per-lane operand multiplexers for scalar broadcast, and a result combiner whose adder tree turns a dot-product step into one instruction. Setting the lane count to one turns the same design into a scalar processor, which gives a like-for-like baseline: same instruction set, same program, same timing rules. A Python golden model, unit and system testbenches and mutation tests verify the design. All six test programs match the model exactly in memory, registers and cycle count for 1, 2, 4 and 8 lanes. On five benchmark kernels, eight lanes need 7.87–$7.92\times$ fewer clock cycles than one. After place-and-route on a Lattice ECP5-25F, the 8-lane unit uses 18% of the logic and closes timing at $33.9\,\mathrm{MHz}$, against $50.8\,\mathrm{MHz}$ for one lane, giving a wall-clock speed-up of $5.3\times$.**

***Index Terms*—SIMD, vector processor, data-level parallelism, FPGA, Verilog, hardware verification.**

I. Introduction

A scalar processor spends most of its effort on control: fetching, decoding and sequencing each instruction. When a program adds two arrays, that effort is repeated for every element, although every element needs the same work. A SIMD processor issues one instruction for a group of elements. The control cost is paid once and the arithmetic is replicated across *lanes*. This is *data-level parallelism*, the form of parallelism behind vector supercomputers, multimedia instruction-set extensions and GPUs [3].

The goal of this project is to design such a unit at a scale that can be built, verified and measured within a semester, and to answer three questions with numbers rather than claims:

1) How close to the ideal $N\times$ does an N-lane unit come, measured in clock cycles?
2) What does each extra lane cost in FPGA logic, multipliers and memory?
3) How much does the wider datapath slow the clock, and what wall-clock speed-up is left?

The Phase I contributions are:

- a synthesizable, lane-parameterised SIMD unit in Verilog with the blocks named in the project brief: vector register file, parallel ALUs, vector control unit, lane multiplexers, result combiner and a scalar interface (Section III);
- a 22-instruction ISA, an assembler and an instruction-level golden model that predicts results and exact cycle counts (Section III-B);
- a layered verification flow with self-checking testbenches, coverage counters and mutation tests (Section III-E);
- a scalar-versus-SIMD comparison on five kernels and an FPGA resource and frequency sweep over 1, 2, 4 and 8 lanes (Section IV).

II. Literature Review

A. Taxonomy and early vector machines

Flynn's taxonomy [1] classifies machines by the number of instruction and data streams. SIMD names the case of one instruction stream acting on many data elements. The first commercially successful SIMD designs were vector supercomputers such as the Cray-1 [2]. Its vector registers and pipelined functional units let one instruction process up to 64 elements, which hid memory latency and avoided the instruction-issue cost of a scalar loop. Hennessy and Patterson [3] trace the same principles from vector machines to subword-SIMD extensions and GPUs.

B. Subword SIMD in general-purpose processors

Intel MMX [4] introduced *packed* data types: one 64-bit register treated as eight bytes or four 16-bit words, with one instruction operating on all of them. Streaming SIMD Extensions (SSE) [5] widened the registers to 128 bits. ARM NEON uses the same width. Two conventions from this line of work are adopted here directly: compare instructions that write all-ones or all-zeros masks, so conditionals need no branch, and a fixed 128-bit register divided into lanes.

C. Scalable vector architectures

VIRAM [6] showed that a vector architecture with lanes scales across embedded media workloads more simply than superscalar or VLIW designs. A compiler could vectorise most kernels automatically. Asanović [7] analysed lane-based vector microprocessors in depth. Lee *et al.* [8] compared MIMD, vector-SIMD, subword-SIMD and SIMT designs and found vector cores more efficient even on fairly irregular kernels. The RISC-V vector extension [9] standardises a length-agnostic vector ISA. Ara [10] implements it with identical lanes, each holding a slice of the vector register file, and reaches over $1\,\mathrm{GHz}$ in 22-nm technology. The lane-replication scheme used in this project follows the same idea at a much smaller scale.

D. Soft vector processors on FPGAs

FPGAs are a natural host for lane-replicated designs because the lane count can be matched to the device. VESPA [11] showed a soft vector processor whose lane count and width can be scaled and customised to each application. VIPERS [12] used a soft vector unit as a general-purpose accelerator for a Nios II processor and reported up to $44\times$ speed-up with 32 lanes. VEGAS [13] replaced the vector register file with a scratchpad and introduced fracturable ALUs. The VectorBlox MXP [14] turned these ideas into a commercial IP core with 1 to over 128 ALUs. These works report speed-up against a separate host processor, so the baseline differs from the vector unit in its microarchitecture as well as its width.

E. Performance limits and tools

Amdahl's law [15] bounds the speed-up of any parallel unit by the fraction of work that stays serial. In a vector loop that fraction is the set-up code, the loop control and the final reduction. Open-source flows now make resource and timing results reproducible: Yosys and nextpnr [16] synthesise and place designs for Lattice ECP5 using documented timing models, and Icarus Verilog [17] simulates them.

F. Gap addressed

Published soft vector processors are large and complex, and they are usually compared against a different scalar core. This project builds a design small enough for every line to be explained and verified. Its scalar baseline is the *same* RTL with one lane, so any measured gain comes only from data-level parallelism.

III. Proposed Solution

A. Overview

Fig. 1 shows the unit. Two parameters set its shape: the number of lanes L (1, 2, 4 or 8) and the element width $W = 16$ bits. With $L = 8$ a vector holds $8 \times 16 = 128$ bits, the width of SSE and NEON. The instruction and data memories sit outside the core, so the same RTL is used both in simulation and on the FPGA (where `fpga_top.v` adds block-RAM memories, switches and LEDs).

B. Instruction set

Instructions are 32 bits: a 6-bit opcode, three 3-bit register fields (`rd`, `ra`, `rb`) and a signed 16-bit immediate. The two top opcode bits select the class:

- **vector–vector**: $v_d[i] = v_a[i] \circ v_b[i]$ for every lane i;
- **vector–scalar**: $v_d[i] = v_a[i] \circ s_b$, the scalar broadcast to all lanes;
- **reduce**: $s_d = \sum_i (v_a[i] \circ v_b[i])$. With $\circ = \times$ this is `VDOT`;
- **memory and control**: `VLD`, `VST`, `ADDI`, `ADD`, `BNE`, `HALT`.

The operation $\circ$ is one of 14 ALU jobs: `ADD`, `SUB`, `AND`, `OR`, `XOR`, three shifts, `MUL` (low half), `SEQ`, `SLT`, `SLTU`, `MIN` and `MAX`. Compare jobs produce all-ones or all-zeros per lane. The vector add kernel, for example, is:

```
    ADDI s1, s0, 0 # word index
    ADDI s2, s0, ITER # passes = N / L
loop: VLD v1, A(s1)
    VLD v2, B(s1)
    VADD v3, v1, v2 # L additions at once
    VST v3, C(s1)
    ADDI s1, s1, 1
    BNE s1, s2, loop
    HALT
```

The same source runs on every lane count; only `ITER` changes.

C. Microarchitecture

a) Vector control unit: Decodes the instruction into control signals and computes the next program counter. Instruction memory is synchronous (block RAM), so the unit sends it the *next* address one cycle early. Taken branches therefore cost nothing. Loads take two cycles: the address goes out in the first, and the returned data is written in the second.

b) Vector register file: Eight $L \cdot W$-bit registers, with two asynchronous read ports and one synchronous write port.

c) Scalar register file (scalar interface): Eight 32-bit registers, with `s0` hard-wired to zero. They hold addresses, loop counters, broadcast values and reduction results. A third, debug read port exposes any register to the host testbench or to LEDs on the board.

d) Lane multiplexers: For each lane, operand B is either that lane's element of v_b or the low W bits of s_b copied to every lane.

e) Parallel ALUs: L instances of a combinational 14-operation lane ALU, created with a `generate` loop and sharing one operation code. Lanes have no connection to each other (no carry crosses a lane boundary).

f) Result combiner: Returns the lane outputs as a vector for write-back. It also sign-extends them to 32 bits and sums them with a $\log_2 L$-level adder tree for the reduce instructions.

D. Timing model

Every instruction except `VLD` completes in one cycle, so a program's run time is

$$\text{cycles} = \text{instructions} + \text{loads}. \tag{1}$$

For a kernel with k set-up instructions, a loop body of b instructions with ℓ loads, and N elements on L lanes,

$$C(L) = k + \frac{N}{L}\,(b + \ell) + 1, \qquad S(L) = \frac{C(1)}{C(L)} \;<\; L. \tag{2}$$

The speed-up S approaches L as N grows; the $k + 1$ serial instructions are the Amdahl term.

E. Verification methodology

Verification works in layers:

1) **Unit testbenches** drive each block with corner cases and seeded random stimulus and compare against reference models written independently of the RTL (for example, shifts as multiplication or division by 2^n). The ALU test performs 225,496 lane checks. It counts true and

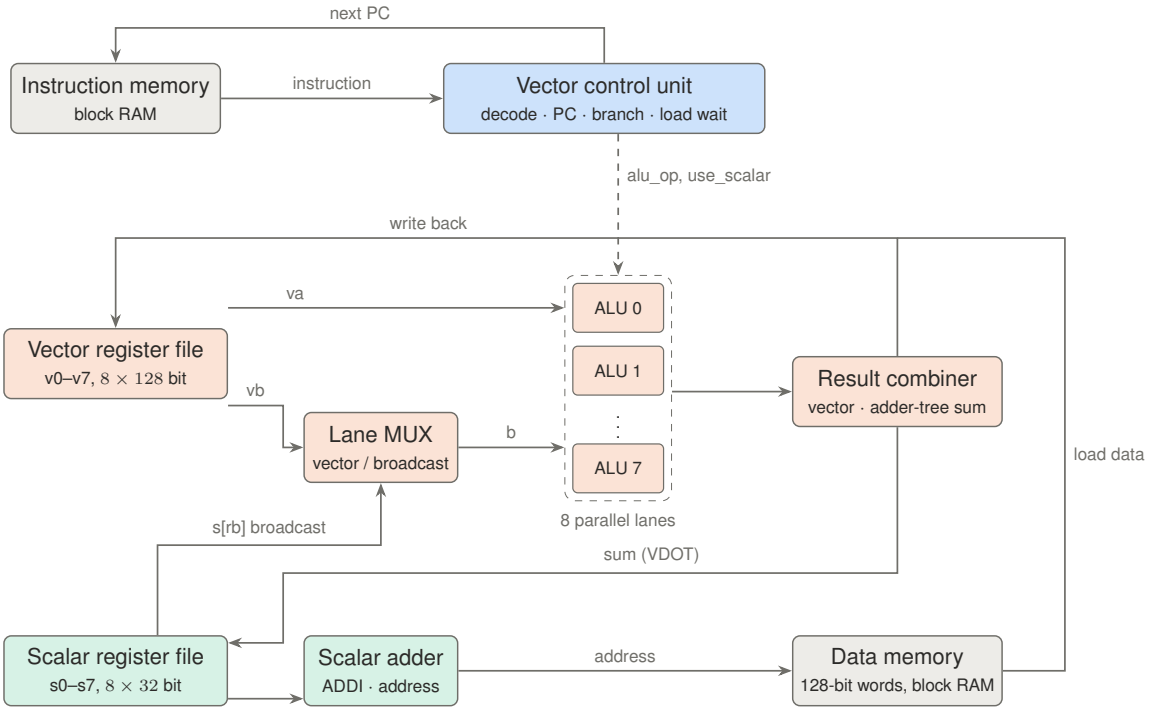


Fig. 1. Block diagram of the SIMD vector processing unit (8 lanes shown). Dashed: control signals.

false outcomes per compare operation and fails on any coverage hole.

2) **Golden model.** A Python instruction-set simulator executes the assembled machine code and predicts the final memory, registers, instruction count and cycle count.
3) **System testbench.** It loads a program and data, runs the core to `HALT`, and compares all 1,024 data-memory words, all scalar registers (through the debug port) and both counters with the golden model.
4) **Program check.** The golden-model result is also compared with plain arithmetic (e.g. $C_i = A_i + B_i$), which catches errors in the programs.
5) **Mutation tests.** Deliberately injected bugs (a logical instead of arithmetic right shift, a mis-wired adder tree, a missing load stall) must make the tests fail.

A sixth program, `isa_test`, executes all 42 vector-ALU and reduce forms on random 16-bit data, so every instruction is exercised on the complete core.

F. Experimental setup

The five benchmark kernels are vector add, SAXPY ($Y = aX + Y$), ReLU, threshold (a branch-free conditional using masks) and dot product, each on $N = 256$ elements with a fixed random seed. The simulator is Icarus Verilog 13; synthesis uses Yosys 0.70 for a Lattice ECP5 LFE5U-25F (speed grade 6), with place-and-route in nextpnr. Each ECP5 build is placed with three seeds and the median frequency is reported. A Yosys estimate for the Xilinx Artix-7 xc7a35t (Basys-3) is also given.

TABLE I
CLOCK CYCLES FOR $N = 256$ ELEMENTS

Kernel	L=1	L=2	L=4	L=8	Speed-up
vadd	2051	1027	515	259	7.92×
saxpy	2308	1156	580	292	7.90×
relu	1539	771	387	195	7.89×
threshold	2053	1029	517	261	7.87×
dot	2052	1028	516	260	7.89×

IV. Prototype Status and Preliminary Results

A. Verification

All unit tests pass, and all 24 system runs (6 programs × 4 lane counts) match the golden model exactly in memory contents, scalar registers, instruction count and cycle count. Every injected mutation was detected. Verilator lint (`-Wall`) reports no warnings for any module; the only two waivers mark bits that are unused on purpose. Continuous integration repeats the lint and tests on every change.

B. Scalar versus SIMD

Table I gives the cycle counts. Every count equals the prediction of (2) exactly, and the speed-up is 1.0–1.6% below the ideal 8×. The gap is the two to four set-up instructions and `HALT`, which are not parallelised. Fig. 2 plots the mean over the five kernels.

C. FPGA resources and frequency

Table II and Fig. 3 summarise the ECP5 results. Each lane adds one hardware multiplier and about 500 logic cells on

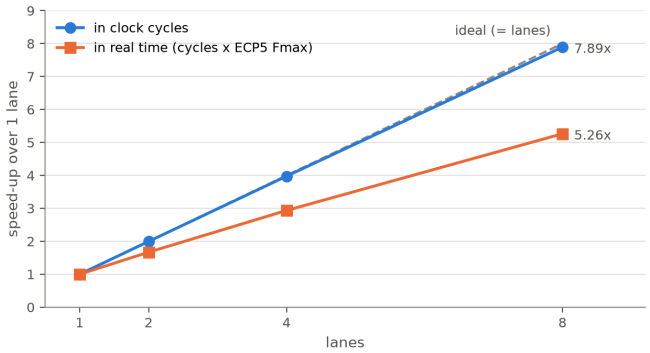


Fig. 2. Speed-up over one lane: in clock cycles (blue) and in wall-clock time once each design's measured ECP5 $F_{\max}$ is applied (orange).

TABLE II
LATTICE ECP5-25F AFTER PLACE-AND-ROUTE

L	Logic cells	Mult.	BRAM	$F_{\max}$ (MHz)
1	921 (3.8%)	1	2	50.8
2	1,604 (6.6%)	2	2	42.5
4	2,851 (11.7%)	4	3	37.5
8	4,418 (18.2%)	8	5	33.9

average. Shared logic (control, scalar registers, memories) is paid once, so eight lanes need $4.8\times$ the logic of one. The register files are mapped to distributed LUT RAM, which keeps the flip-flop count low. On the Artix-7, Yosys estimates 3,698 LUTs (18%) and 9 DSP slices for eight lanes.

D. Discussion

The maximum frequency falls from $50.8\,\mathrm{MHz}$ to $33.9\,\mathrm{MHz}$ as the lanes go from one to eight. Because the core executes each instruction in a single cycle, its critical path crosses the whole datapath: block-RAM output, decode, register read, multiplier, adder tree and scalar write-back. Wider designs also spread across more of the device, which lengthens routing. The wall-clock speed-up at eight lanes is therefore $7.89 \times 33.9/50.8 = 5.3\times$, not $7.9\times$. The result confirms the main promise of SIMD: near-linear gains in work per instruction. It also shows that a wider datapath must be pipelined to keep its clock rate. On a small Lattice iCE40 UP5K (5,280 cells), the 8-lane design did not fit even with its eight DSP blocks in use. A mid-range device is the practical minimum.

V. Individual Contributions

Mahir defined the instruction format and the block interfaces that the rest of the team built against. He wrote the vector control unit, with its early instruction fetch and two-cycle load, and integrated the core. He also wrote the system testbench that compares memory, registers and cycle counts with the golden model, and set up the repository, continuous integration and documentation.

Krishiv Gupta designed the lane ALU and its `generate`-based array, including the handling of signed and unsigned compares and arithmetic shifts. His testbench checks 225,496 lane results against an independent integer model and proves that no carry crosses a lane boundary.

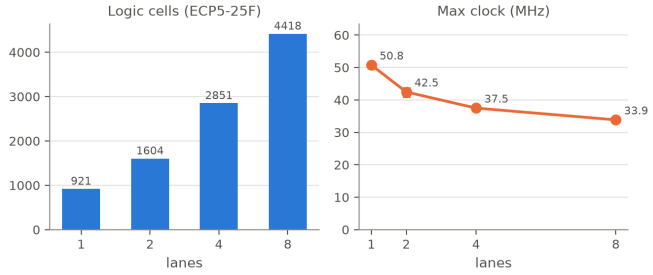


Fig. 3. ECP5-25F logic cells and maximum clock frequency against lane count (median of three placement seeds; bars show the range).

TABLE III
WORK SPLIT (GITHUB HISTORY SHOWS EACH MEMBER'S COMMITS)

Member	Contribution
Mahir (lead)	Architecture and ISA specification; vector control unit; core integration; system testbench; CI; documentation; report
Krishiv Gupta	Lane ALU (14 operations) and parallel ALU array; ALU testbench with corner, random and coverage checks
Abhiman Singh	Vector and scalar register files with debug port; register-file testbench
Jayadeep	Lane multiplexers and result combiner (adder tree); combiner testbench for 1–8 lanes; FPGA wrapper and synthesis flow (ECP5, Artix-7, Vivado script)
Ashish Kushwaha	Assembler, golden model and six workload programs; workload runner; scalar-versus-SIMD analysis and figures

Abhiman Singh designed the two register files: two read ports and one write port each, a hard-wired zero register, and the debug port that forms the scalar interface. His testbench covers read-during-write behaviour and 4,000 random cycles against a shadow model.

Jayadeep designed the lane multiplexers and the result combiner's $\log_2 L$ adder tree, verified for every lane count. He built the FPGA wrapper and the synthesis flow that produced the resource and frequency results, including the Vivado script and Basys-3 constraints for the lab.

Ashish Kushwaha wrote the assembler, the instruction-level golden model and the six workload programs. He also wrote the runner that checks each program against plain arithmetic and against the RTL, and the scalar-versus-SIMD analysis and figures.

VI. Conclusion and Phase II Plan

The Phase I prototype is complete and verified. An eight-lane, 128-bit SIMD unit executes a 22-instruction ISA, matches its golden model cycle for cycle, and reaches a $7.9\times$ cycle speed-up and a $5.3\times$ wall-clock speed-up over its own scalar configuration on a mid-range FPGA. Phase II will (i) pipeline the core to raise $F_{\max}$ and recover wall-clock speed-up; (ii) measure exact Artix-7 resources and timing in

Vivado and run the design on a Basys-3 board; (iii) add predicated (masked) execution and saturating arithmetic; (iv) study speed-up against problem size N to quantify the Amdahl term; and (v) compare power estimates per operation for scalar and SIMD builds.

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